

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 03

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Abinesh H (192424387) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Criteria	Conceptual Understanding	Accuracy of Answers	Application of Knowledge	Completeness of Response	Clarity & Presentation	Total
Weightage	30%	20%	20%	15%	15%	
Q1						
Q2						

Signature of the student: Abinesh H (192424387)

Total Marks:

Annexure A

Short Answer Test

Question 1: Explain how Bayes' Theorem can be applied to the proposed probabilistic disease diagnosis system.

Answer

Medical diagnosis involves uncertainty because the same symptom can be associated with different diseases. For example, fever and cough may occur in Influenza, Pneumonia, COVID-19, or even the Common Cold. Therefore, a diagnostic system should not simply classify a patient as having or not having a disease. Instead, it should calculate the probability of each possible disease based on the available evidence.

Bayes' Theorem provides a mathematical method for updating the probability of a disease when new patient information becomes available.

1. Bayes' Theorem

The general form of Bayes' Theorem is:

Where:

- = Posterior probability — probability that the patient has disease after observing evidence .
- = Likelihood — probability of observing the patient's evidence if the patient has disease .
- = Prior probability — probability of the disease before considering the current patient's evidence.
- = Evidence probability — overall probability of observing the given evidence.

The main purpose of Bayes' Theorem is therefore to update prior medical knowledge using new patient-specific evidence.

2. Application to the Given Case Study

The hospital considers four possible diseases:

1. Influenza
2. Pneumonia
3. COVID-19
4. Common Cold

Suppose a patient arrives with the following symptoms:

- Fever
- Cough
- Shortness of breath
- Fatigue

The system initially has prior probabilities for each disease based on historical hospital data.

For example, assume:

Disease	Prior Probability
Influenza	0.30
Pneumonia	0.20
COVID-19	0.25
Common Cold	0.25

The system then receives the patient's symptoms as new evidence.

Let the complete evidence be:

For each disease, the system determines how likely this evidence is.

For example:

represents the probability of observing these symptoms in a patient who has COVID-19.

Similarly:

represents the probability of observing the same evidence in a patient with Pneumonia.

Bayes' Theorem is then applied to each disease.

3. Posterior Probability Calculation

For COVID-19:

For Pneumonia:

The same process is performed for Influenza and Common Cold.

Since the denominator is common to all four diseases, the system can calculate the relative probabilities and normalize them so that their total equals 1.

For the case study, suppose the final estimated probabilities are:

Disease	Posterior Probability
Influenza	0.20
Pneumonia	0.35

COVID-19	0.40
Common Cold	0.05
Total	1.00

In percentage form:

- Influenza = 20%
- Pneumonia = 35%
- COVID-19 = 40%
- Common Cold = 5%

Therefore:

which is the highest probability.

4. Updating the Diagnosis with New Evidence

One of the major advantages of Bayes' Theorem is that the diagnosis can be updated whenever new evidence becomes available.

For example, suppose the patient initially has fever and cough. The system may calculate:

- Influenza = 30%
- Pneumonia = 25%
- COVID-19 = 35%
- Common Cold = 10%

Later, the patient undergoes a chest X-ray and the result shows findings strongly associated with Pneumonia.

The system can use this new evidence and recalculate:

The probabilities may change significantly.

This means the diagnostic system is dynamic rather than static. Every new symptom, laboratory result, or medical observation can modify the estimated disease probabilities.

5. Role of Patient Information

The proposed system can consider several types of patient information:

Patient Information	Role in Diagnosis
Fever	Helps distinguish between respiratory infections
Cough	Provides evidence for respiratory diseases

Shortness of breath	May indicate more serious respiratory conditions
Sore throat	Can occur in several respiratory infections
Fatigue	Provides additional diagnostic evidence
Age	Can influence disease risk
Smoking history	Can influence respiratory disease probability
Previous respiratory disease	Provides additional risk information
Oxygen saturation	Important indicator of respiratory condition
Chest X-ray	Provides clinical evidence for conditions such as Pneumonia

Bayes' Theorem allows these different pieces of evidence to be incorporated into the probability calculation.

6. Learning Probabilities from Historical Data

The probabilities used by the system can be learned from historical patient records.

For example, if a hospital has records of thousands of patients, the system can estimate:

from the proportion of patients diagnosed with COVID-19.

It can also estimate probabilities such as:

These learned probabilities make the diagnostic system more representative of actual clinical data.

As more reliable patient data becomes available, the probability estimates can be updated and improved.

7. Advantages of Using Bayes' Theorem

The use of Bayes' Theorem provides several important advantages:

1. **Handles uncertainty:** Medical symptoms do not always lead to a definite diagnosis.
2. **Uses multiple sources of evidence:** Symptoms, medical history, laboratory results, and imaging findings can be incorporated.
3. **Updates probabilities:** The diagnosis can change when new evidence is obtained.
4. **Provides interpretable results:** Doctors can see the probability associated with each disease.

5. **Supports clinical decision-making:** It helps doctors identify which conditions are more likely.
6. **Uses historical knowledge:** Previous patient data can be used to establish prior probabilities.
7. **Avoids absolute conclusions:** The system reports likelihood rather than claiming certainty.

8. Final Interpretation

In the given example, the final probabilities are:

Since COVID-19 has the highest probability, the system would identify it as the most probable condition.

However, a 40% probability does not mean that the patient definitely has COVID-19. It only means that, based on the available evidence and learned probabilities, COVID-19 is currently the most likely among the considered conditions.

A doctor should therefore use the result along with clinical examination, laboratory tests, imaging, and other relevant medical information before making a final diagnosis.

Conclusion

Bayes' Theorem provides a systematic mathematical framework for probabilistic medical diagnosis. It combines prior knowledge with patient-specific evidence to calculate updated disease probabilities. In the proposed hospital system, it enables continuous updating of diagnostic predictions as new information becomes available. Thus, the system can serve as an effective clinical decision-support tool, while the final medical decision remains with qualified healthcare professionals.

Question 2: Explain the role of Naïve Bayes and Bayesian Belief Networks (BBNs) in the proposed disease diagnosis system.

Answer

The proposed diagnostic system can use Naïve Bayes and Bayesian Belief Networks (BBNs) to perform probabilistic reasoning. Both methods are based on probability theory, but they represent relationships among medical variables differently.

1. Naïve Bayes

Naïve Bayes is a probabilistic classification method based on Bayes' Theorem. It assumes that the observed features are conditionally independent given the disease.

For multiple symptoms:

For example, for COVID-19:

The same calculation can be performed for Influenza, Pneumonia, and Common Cold.

The disease having the highest calculated probability is selected as the most probable disease.

Advantages of Naïve Bayes

- Simple to implement.
- Computationally efficient.
- Fast even with large datasets.
- Requires relatively less training data.
- Produces probabilistic results.
- Useful as a baseline diagnostic model.

Limitation

The main limitation is its conditional independence assumption. In real medical situations, symptoms may not be completely independent. For example, fever, fatigue, and increased heart rate may be related.

2. Bayesian Belief Networks

A Bayesian Belief Network is a graphical probabilistic model that represents relationships between variables using nodes and directed connections.

For the given medical application, a simplified network could be represented conceptually as:

Age → Disease ← Smoking History

Previous Respiratory Disease → Disease

Disease → Fever

Disease → Cough

Disease → Shortness of Breath

Disease → Oxygen Saturation

Disease → Chest X-ray Findings

Here, the Disease node represents the possible conditions, while symptoms, medical history, and test results provide evidence.

Unlike Naïve Bayes, a BBN can represent dependencies between variables.

For example:

- Smoking history can affect respiratory disease risk.
- Previous respiratory disease can influence disease severity.
- Pneumonia can lead to low oxygen saturation.
- Pneumonia can produce characteristic chest X-ray findings.

When new evidence is entered into the network, the probabilities of related diseases are updated.

3. Comparison Between Naïve Bayes and BBN

Feature	Naïve Bayes	Bayesian Belief Network
Structure	Probabilistic classifier	Graphical probabilistic model
Feature assumption	Assumes conditional independence	Represents dependencies
Complexity	Low	Higher
Processing speed	Very fast	Relatively more computationally intensive
Interpretability	Simple	High
Medical relationships	Limited representation	Can represent complex relationships
Suitable application	Fast baseline diagnosis	Detailed clinical reasoning

4. Role of Historical Patient Data

Historical patient records can be used to learn the probabilities required by both models.

For Naïve Bayes, the system can learn probabilities such as:

and

For a BBN, the system can learn conditional probability tables representing relationships between connected variables.

Therefore, as the hospital obtains more reliable patient data, the model can potentially improve its probability estimates.

5. Overall Role in the Proposed System

The two approaches can complement each other.

Naïve Bayes can provide a fast initial probability estimate for the four diseases. A Bayesian Belief Network can then provide a richer representation of relationships involving age, smoking history, previous respiratory disease, symptoms, oxygen saturation, and X-ray findings.

The overall system can therefore follow this general process:

Patient Data → Probability Calculation → Disease Probability Estimation → Ranking of Diseases → Doctor's Clinical Evaluation

The system may produce an output such as:

Disease	Probability	Ranking
COVID-19	40%	1
Pneumonia	35%	2
Influenza	20%	3
Common Cold	5%	4

This gives the doctor a clear view of the relative likelihood of each condition.

Conclusion

Naïve Bayes provides a simple, fast, and efficient baseline approach for probabilistic disease classification, whereas Bayesian Belief Networks provide a more flexible framework for modeling dependencies among medical variables. By combining these approaches with probabilities learned from historical patient data, the proposed system can provide meaningful probabilistic decision support. The system should be treated as an assistive diagnostic tool and not as a replacement for professional medical judgment.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand